

VoxNeuro: Leakage-Safe Parkinson’s Screening Support via Multi-View Grassmannian Speech Analysis

Laiba Khan^{1*} and Manas Waghe²

¹ Lexington High School, Lexington, MA 02421, USA; 27stu469@lexingtonma.org

² Worcester Polytechnic Institute, Worcester, MA 01609, USA; mywaghe@wpi.edu

* Correspondence: 27stu469@lexingtonma.org

Abstract: Parkinson’s disease (PD) can alter phonation and articulation, making repeated speech a candidate digital biomarker. Public speech datasets contain several recordings per participant, and record-wise cross-validation can place one individual in both training and test sets, so reported scores partly measure recognition of an observed person rather than generalization to an unseen participant. VoxNeuro treats the subject as the statistical observation. Each subject’s three recordings define a rank-three Grassmann subspace and a Euclidean mean–dispersion vector; Gaussian kernels on the two views are fused into a positive-semidefinite kernel for support vector classification, and minority subjects are synthesized along Grassmann geodesics. Under a fixed five-fold subject-disjoint protocol, the fused model reached balanced accuracies of 0.850 (UCI-489) and 0.731 (PD-252) and improved macro-F1 over a matched Euclidean radial-basis-function ablation in 23 of 24 augmentation seeds and 13 of 20 outer partitions on PD-252; nested selection of the fusion weight made its balanced-accuracy gain consistent (20 of 24 seeds), and the complete geometric pipeline exceeded a fully Euclidean kernel pipeline in macro-F1 in 24 of 24 seeds and 18 of 20 partitions. These results support subject-disjoint evaluation with explicit robustness reporting and show a modest, consistent benefit of geometry on the imbalanced cohort.

Keywords: Parkinson’s disease; speech biomarker; Grassmann manifold; kernel fusion; imbalanced learning; subject-disjoint cross-validation; support vector machine; digital health

1. Introduction

Parkinson’s disease (PD) is a progressive neurodegenerative disorder involving nigrostriatal dopaminergic loss, basal ganglia circuit dysfunction, and heterogeneous motor and non-motor manifestations [1–3]. Its diagnosis remains clinical [4,5], and its global burden has grown faster than that of any other neurological disorder [6,7]. Disordered speech is among the most common motor consequences: dysarthria, hypophonia, and related vocal-tract dysfunctions affect the large majority of patients over the disease course [8,9]. Quantitative acoustic analysis detects these changes even in early and untreated disease [10]. Nonlinear dysphonia measures separate disordered from healthy voices [11], and repeated noninvasive speech tests have supported research on low-burden remote monitoring and progression tracking [12–14]. Automated speech analysis has revealed prodromal changes in at-risk cohorts [15]. Machine-learning approaches to voice-based PD detection have accordingly expanded rapidly, as documented by recent systematic reviews [16,17]. A digital biomarker, however, is clinically credible only when its evaluation matches its intended unit of use: the patient.

Public PD speech datasets often contain multiple recordings per participant. When recordings are shuffled independently, subject-specific anatomical and acquisition signatures can occur on both sides of a record-wise split, and the evaluation then estimates recognition of another recording from a previously observed person rather than generalization to an unseen person. This failure mode is a specific instance of data leakage, a recognized driver of over-optimistic results across applied machine learning [18,19]. This effect has been demonstrated directly in clinical prediction: record-wise cross-validation can inflate accuracy substantially relative to subject-wise evaluation of the very same pipeline [20], a finding that prompted an extended methodological discussion [21]. Subject characteristics can also drive apparent performance even when splits look innocuous [22]. Related biases arise whenever model selection and error estimation share data [23,24], and cross-validation on small samples carries wide error bars even under favorable conditions [25–27]. The UCI-489 corpus analyzed here contains

three dependent replications from each of 80 subjects [28]; replication-aware analysis has been proposed for exactly this cohort [29].

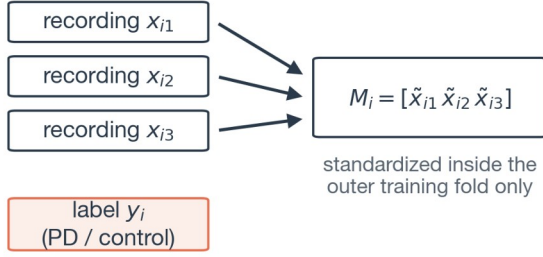
Methodologically, three research strands converge in this study. First, engineered speech descriptors: dysphonia, cepstral, wavelet, and tunable Q-factor wavelet transform (TQWT) descriptor families have been used to characterize PD-related voice changes [12,30,31]. Second, subspace geometry: collections of related observations can be represented as points on the Grassmann manifold $\text{Gr}(r, d)$, whose quotient structure removes arbitrary basis choices [32–34]; this representation underpins subspace-based learning in computer vision and pattern recognition [35,36], and kernels on such manifolds require care because Gaussian functions of arbitrary manifold distances are not automatically positive semidefinite (PSD) [37]. Third, kernel fusion: reproducing-kernel Hilbert space (RKHS) theory [38–40] and multiple-kernel learning [41–43] combine heterogeneous similarities while preserving convex support vector optimization. Class imbalance, finally, is routinely addressed by synthetic oversampling [44–46], whose behavior in high-dimensional feature spaces warrants caution [47]. This study combines these strands under an evaluation protocol that treats the patient as the statistical unit.

Deep architectures, including convolutional, recurrent, and transformer-based models, are increasingly applied to voice-based PD detection and scale well when large corpora are available [16]. Smartphone-scale collection efforts such as mPower point in that direction [48,49], and technology roadmaps for PD anticipate exactly this kind of remote, high-frequency assessment [50–52]. The cohorts relevant to replicated-phonation screening, however, typically contain tens to a few hundred subjects. In this regime, the present study deliberately combines structured geometric priors with a convex, auditable classical learner instead of high-capacity end-to-end models: each component (representation, kernel, solver, and resampling step) has an explicit mathematical description that can be verified directly against its implementation.

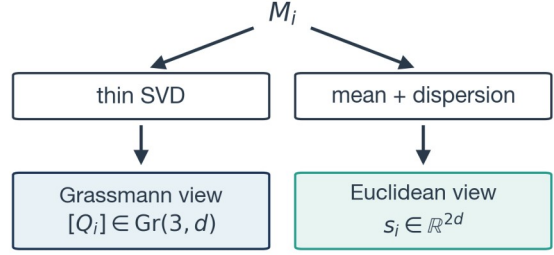
VoxNeuro aligns the mathematical unit of analysis with the clinical unit of interest: each subject is represented by the complete set of repeated measurements. A fixed-rank representation based on singular value decomposition (SVD) encodes the joint orientation of the repetitions, while a Euclidean summary preserves feature-space location and within-subject variability. A positive-semidefinite direct-sum kernel combines the two views, and evaluation is subject-disjoint throughout (Figure 1). For imbalanced data, synthetic subjects are generated along manifold geodesics rather than by interpolating individual recording rows. The study had two objectives: (i) to formalize why record-wise and subject-disjoint evaluation estimate different risks; and (ii) to assess whether adding the Grassmann kernel to a matched nonlinear Euclidean model improves subject-level classification.

The principal contributions are as follows: (i) a formal subject-disjoint risk definition together with a quantitative account of the optimistic bias induced by record-wise splitting; (ii) a basis-invariant Grassmann representation of repeated speech; (iii) provably positive-semidefinite fusion of geometric and Euclidean kernels; (iv) geodesic minority oversampling coupled to the Euclidean view; and (v) a separate research-stage web application for guided speech capture and longitudinal result visualization, not used in the present analyses. One fixed protocol is applied to both cohorts with a common comparator set, including a matched nonlinear Euclidean ablation with a radial basis function (RBF) kernel that isolates the contribution of the Grassmann kernel term, and robustness to the augmentation seed, the outer subject partition, the subspace rank, and the augmentation family is reported.

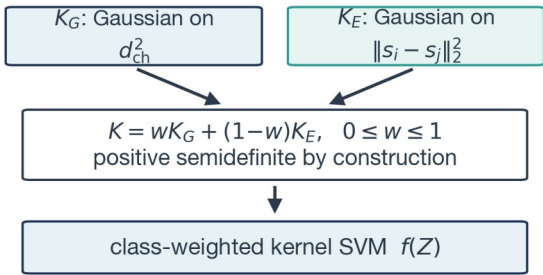
A Subject-level observation



B Two complementary views



C PSD kernel fusion



D Leakage-safe evaluation

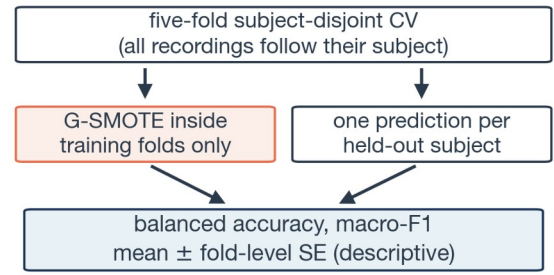


Figure 1. Schematic overview of the VoxNeuro framework. (A) Each subject contributes three repeated recordings and one diagnostic label; the recordings are standardized with statistics estimated inside the outer training fold only and stacked into a matrix M_i . (B) A thin singular value decomposition (SVD) yields the Grassmann view $[Q_i]$, while coordinate-wise mean and dispersion yield the Euclidean view s_i . (C) Gaussian kernels on the two views are combined into a positive-semidefinite (PSD) fused kernel feeding a class-weighted support vector machine (SVM). (D) Evaluation is five-fold subject-disjoint cross-validation (CV); conditional Grassmann-geodesic synthetic minority oversampling (G-SMOTE) is activated only inside imbalanced training folds, and fold scores are summarized as mean $\pm$ standard error (SE). PD, Parkinson’s disease.

2. Materials and Methods

2.1. Datasets

Two public, de-identified cohorts were analyzed (Table 1). The first, denoted UCI-489, comprises 80 subjects (40 PD, 40 control) with three recordings of sustained /a/ per subject, each recording described by 44 replicated acoustic descriptors and one dataset-provided binary sex/gender variable [28,29]. The second, denoted PD-252 [53], comprises 252 subjects (188 PD, 64 control) with three recordings per subject, described by 752 descriptors spanning baseline acoustic, vocal/time-frequency, Mel-frequency cepstral coefficient (MFCC) [54], wavelet, TQWT [55], and intrinsic mode function/empirical mode decomposition (IMF/EMD) [56] families [30,31]. It also includes one dataset-provided binary sex/gender variable, which constitutes the demographic family of the permutation analysis in Section 3.4. The coded variable was included as a fixed model input and was never used to construct folds. The cohorts were selected as complementary evaluation regimes for the same repeated-measurement protocol: UCI-489 evaluates the representation in a balanced, lower-dimensional setting without synthetic augmentation, whereas PD-252 evaluates the same representation in a larger, high-dimensional, imbalanced setting in which the prespecified imbalance-handling branch is activated. This study was a secondary computational analysis of public data and involved no participant recruitment.

Table 1. Dataset regimes and evaluation design.

Attribute	UCI-489	PD-252
Subjects	80	252
PD / control	40 / 40	188 / 64
Recordings per subject	3	3
Modeled variables	45	753
Acoustic regime	Sustained / a/	Multi-family engineered features
Oversampling	Inactive	Minority control class only
Outer evaluation	Five-fold subject-disjoint cross-validation	

2.2. Subject-Disjoint Problem Formulation

Let $R_i = \{(x_{ij}, y_i) : j = 1, \dots, m_i\}$ denote all recordings from subject i ; Table 2 summarizes the notation used throughout. For outer fold k , let I_k^{tr} and I_k^{te} be the training and held-out subject-index sets. Leakage-safe evaluation requires

$$I_k^{\text{tr}} \cap I_k^{\text{te}} = \emptyset, I_k^{\text{tr}} \cup I_k^{\text{te}} = \{1, \dots, n\}. \quad (1)$$

All recordings in R_i are assigned to the same partition as subject i (Figure 2A,B). For a pointwise classification loss ℓ , the fold risk is evaluated once per held-out subject,

$$\hat{R}_k(f) = \frac{1}{|I_k^{\text{te}}|} \sum_{i \in I_k^{\text{te}}} \ell(f(Z_i), y_i), \quad (2)$$

where Z_i is the complete subject-level representation. Equations (1) and (2) define the intended estimand, generalization to an unseen person rather than to an unseen row from a known person; the reported balanced accuracy and macro-F1 are fold-level aggregate metrics defined in Equation (15).

Table 2. Principal notation.

Symbol	Meaning
$x_{ij} \in \mathbb{R}^d$	feature vector of recording j of subject i
R_i, y_i	all recordings of subject i and its diagnostic label
$I_k^{\text{tr}}, I_k^{\text{te}}$	training and held-out subject-index sets of outer fold k
M_i	standardized $d \times 3$ recording matrix of subject i
$Q_i, [Q_i]$	orthonormal basis from the thin SVD and its subspace class
$\text{Gr}(r, d), \text{St}(r, d)$	Grassmann and Stiefel manifolds of rank r in $\mathbb{R}^d$
$P_i = Q_i Q_i^\top$	orthogonal projector representing $[Q_i]$
$\theta_{ij, q'}, d_{\text{ch}}$	principal angles and chordal distance between subspaces
$s_i \in \mathbb{R}^{2d}$	Euclidean mean–dispersion summary of subject i
$K_G, K_E, K; w$	Grassmann, Euclidean, and fused kernels; fusion weight
γ_G, γ_E	Gaussian kernel coefficients (inverse squared bandwidths) of the two views
$Q_{\text{syn}}(t), s_{\text{syn}}(t)$	geodesic and Euclidean interpolants of G-SMOTE
C_{y_i}	class-weighted SVM box constraint for class y_i
$F, \bar{m}, \text{SE}$	number of outer folds, fold-mean metric, standard error
I_{gk}, π_g	fold-level permutation importance and permutation of family g
$h, A, R_{\text{seen}}, R_{\text{new}}, R_{\text{row}}$	record-level classifier, leakage event, and associated risks

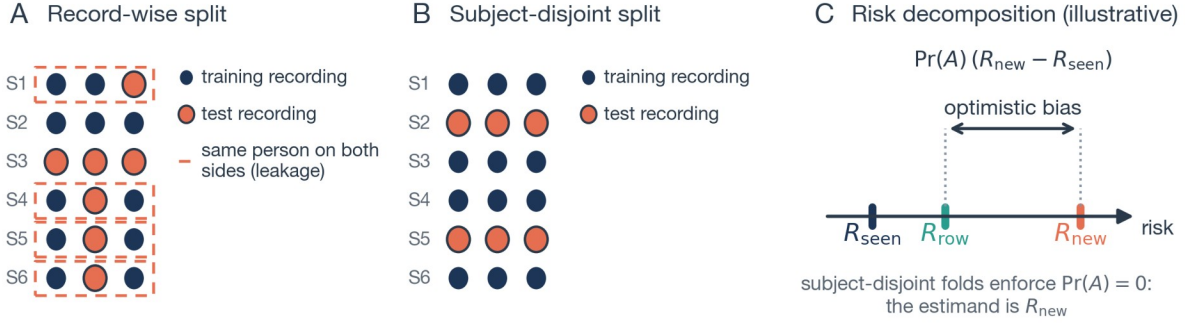


Figure 2. Schematic illustration of evaluation leakage with repeated recordings. (A) A record-wise split assigns individual recordings to training or test sets, so several subjects (dashed outlines) appear on both sides and the test set is not independent of training identities. (B) A subject-disjoint split assigns all recordings of a subject jointly. (C) Consequence of Equation (18): the record-wise estimate R_{row} interpolates between the easier seen-person risk R_{seen} and the deployment-relevant unseen-person risk R_{new} , drawn for an illustrative $\Pr(A) = 0.65$; the arrow marks the optimistic bias of magnitude $\Pr(A)(R_{\text{new}} - R_{\text{seen}})$.

2.3. Standardization and Subject Representation

For subject i , the three acoustic vectors $x_{ij} \in \mathbb{R}^d$ are standardized using a mean vector μ_k and a scale vector estimated exclusively from the recordings of training subjects in outer fold k . Under the implemented standardization procedure, a zero-variance training coordinate is assigned unit scale:

$$\begin{aligned} \tilde{x}_{ij} &= D_k^{-1}(x_{ij} - \mu_k), D_k = \text{diag}(\sigma_k^*), \\ M_i &= [\tilde{x}_{i1} \tilde{x}_{i2} \tilde{x}_{i3}], \end{aligned} \quad (3)$$

where σ_k^* denotes the training-fold scale vector after zero-variance handling. A thin SVD [57] gives

$$\begin{aligned} M_i &= U_i \Sigma_i V_i^T, Q_i = U_i(:, 1:r), \\ Q_i^T Q_i &= I_r, [Q_i] \in \text{Gr}(r, d). \end{aligned} \quad (4)$$

Throughout, the rank is fixed at $r = 3$, matching the three repeated recordings per subject. The basis is not unique: Q_i and $Q_i O$, with O in the orthogonal group $O(r)$, represent the same subspace. This quotient structure removes coordinate choices within the retained SVD basis (Figure 3A). In the official releases, one subject per cohort (CONT-36 in UCI-489; subject 37 in PD-252) contains byte-identical repeated recordings, so its matrix has numerical rank two. For such a subject, Equation (4) retains the orthonormal completion supplied by the SVD as the third basis vector, and the reference implementation accepts it only under an explicit opt-in flag.

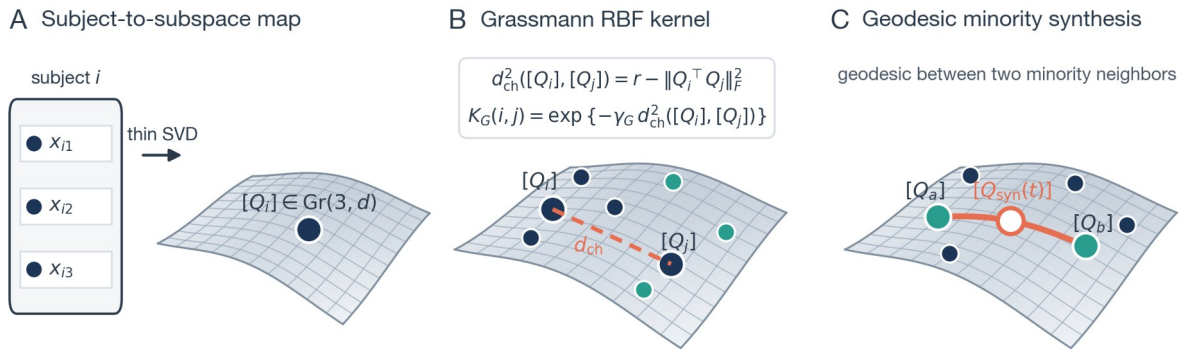


Figure 3. Geometric interpretation of the VoxNeuro representation (schematic; the curved sheet stands for the Grassmann manifold). (A) The three repeated recordings of a subject are mapped by a thin SVD to a single point $[Q_i]$ on $\text{Gr}(3, d)$. (B) The chordal distance d_{ch} between two subject subspaces (dashed chord), whose square enters

the displayed Gaussian Grassmann radial-basis-function kernel, compares subjects independently of the arbitrary orthonormal basis selected for each subspace; navy and teal points denote subjects of the two classes. (C) Schematic of minority synthesis along the Grassmann geodesic joining two minority neighbors, rather than row-wise Euclidean interpolation.

2.4. Principal Angles and Projection Geometry

Principal angles between two subject subspaces (Figure 4A) are defined by [32,58]

$$Q_i^\top Q_j = A_{ij} \cos(\Theta_{ij}) B_{ij}^\top, \quad \Theta_{ij} = \text{diag}(\theta_{ij,1}, \dots, \theta_{ij,r}), \quad (5)$$

where A_{ij} and B_{ij} are orthogonal. With the projector $P_i = Q_i Q_i^\top$, the squared chordal distance is given by [33,59]

$$\begin{aligned} d_{\text{ch}}^2([Q_i], [Q_j]) &= \frac{1}{2} \|P_i - P_j\|_F^2 \\ &= r - \|Q_i^\top Q_j\|_F^2 = \sum_{q=1}^r \sin^2 \theta_{ij,q}. \end{aligned} \quad (6)$$

Because $(Q_i O)(Q_i O)^\top = P_i$, the distance is basis invariant (Figure 4B). Vectorizing P_i maps the chordal distance to a Euclidean distance in projector space, up to a constant factor $\sqrt{2}$ absorbed by the kernel coefficient, thereby providing a Euclidean embedding for the Gaussian Grassmann kernel.

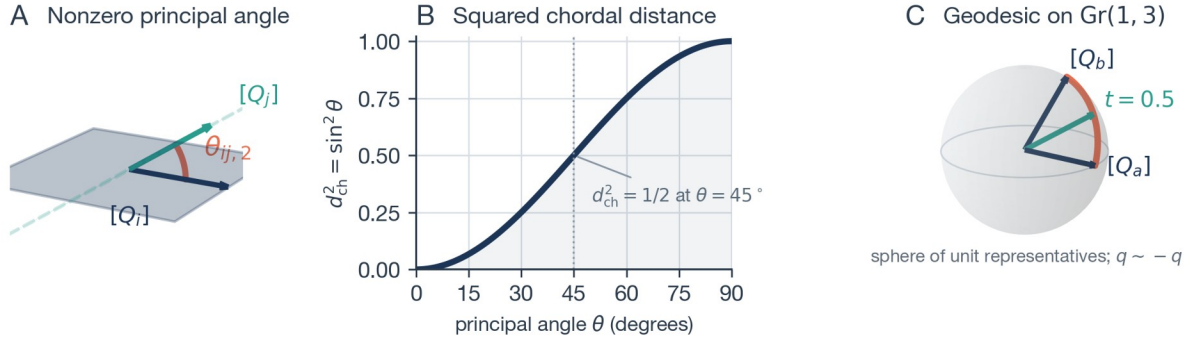


Figure 4. The geometry behind Equations (5), (6), (11), and (12), drawn for illustrative toy subspaces. (A) Two planes in $\mathbb{R}^3$ that share one direction, so the first principal angle is zero and the drawn angle is the second principal angle $\theta_{ij,2}$. (B) Exact rank-one relation between a principal angle and the squared chordal distance, $d_{\text{ch}}^2 = \sin^2 \theta$. (C) A geodesic path computed with Equations (11) and (12) between two one-dimensional subspaces $[Q_a]$ and $[Q_b]$; every interpolated frame in the drawn path satisfies $Q_{\text{syn}}(t)^\top Q_{\text{syn}}(t) = I_r$ to machine precision, consistent with Proposition 2.

2.5. Complementary Euclidean View

The complementary Euclidean view retains the coordinate-wise mean and dispersion:

$$\begin{aligned} \mu_i &= \frac{1}{3} \sum_{j=1}^3 \tilde{x}_{ij}, \sigma_i = \left[\frac{1}{3} \sum_{j=1}^3 (\tilde{x}_{ij} - \mu_i)^{\odot 2} \right]^{1/2}, \\ s_i &= [\mu_i^\top \sigma_i^\top]^\top \in \mathbb{R}^{2d}, \end{aligned} \quad (7)$$

where $(\cdot)^{\odot 2}$ and the square root are applied elementwise. Thus $[Q_i]$ encodes the orientation of the span of the standardized repetitions (M_i is not centered within subject, so this span generally includes the mean direction), whereas s_i retains coordinate-level location and dispersion.

2.6. Direct-Sum Kernel and Guarantee of Positive Semidefiniteness

Let ϕ_G and ϕ_E be feature maps for Gaussian kernels on the projection and Euclidean views [38,39]. With fusion weight w , the joint representation is the Hilbert direct sum

$$\Phi(Z_i) = (\sqrt{w}\phi_G([Q_i]), \sqrt{1-w}\phi_E(s_i)), \quad 0 \leq w \leq 1, \quad (8)$$

which induces, for kernel coefficients $\gamma_G, \gamma_E > 0$ (inverse squared bandwidths),

$$\begin{aligned} K_G(i, j) &= \exp\{-\gamma_G d_{\text{ch}}^2([Q_i], [Q_j])\}, \\ K_E(i, j) &= \exp\{-\gamma_E \|s_i - s_j\|_2^2\}, \\ K(i, j) &= w K_G(i, j) + (1-w) K_E(i, j). \end{aligned} \quad (9)$$

Proposition 1 (*basis invariance and positive semidefiniteness*). The kernel in Equation (9) is invariant to $Q_i \mapsto Q_i O$ and is PSD. Basis invariance follows from the projector representation in Equation (6). Each Gaussian block is PSD on its Euclidean embedding, and a nonnegative weighted sum of PSD kernels is PSD [39,41]; hence, for every $c \in \mathbb{R}^n$,

$$c^T K c = w c^T K_G c + (1-w) c^T K_E c \geq 0. \quad (10)$$

Equivalently, $K(i, j) = \langle \Phi(Z_i), \Phi(Z_j) \rangle$ in the direct-sum reproducing-kernel Hilbert space. $\square$ The projector embedding is essential: Gaussian kernels built on other manifold distances, such as the geodesic arc length, are not PSD for all bandwidths [37], whereas the chordal construction used here inherits positive semidefiniteness from its explicit Euclidean embedding.

2.7. Grassmann-Geodesic Synthetic Minority Oversampling

This project-specific manifold adaptation is referred to as Grassmann-geodesic SMOTE (G-SMOTE); the abbreviation does not denote the separate Geometric SMOTE algorithm of Euclidean imbalanced learning [60]. Following the synthetic-neighbor principle of standard SMOTE [44] and its descendants [45,46,61], for minority subjects a and b ,

$$\begin{aligned} Q_a^T Q_b &= U \cos(\Theta) V^T, \\ W_{ab} &= (Q_b V - Q_a U \cos \Theta) (\sin \Theta)^\dagger, \end{aligned} \quad (11)$$

where $\dagger$ denotes the Moore–Penrose pseudoinverse. A shortest geodesic and its coupled Euclidean interpolation are defined as follows (Figure 4C):

$$\begin{aligned} Q_{\text{syn}}(t) &= [Q_a U \cos(t\Theta) + W_{ab} \sin(t\Theta)] V^T, \\ s_{\text{syn}}(t) &= s_a + t(s_b - s_a), \quad t \sim U(0, 1). \end{aligned} \quad (12)$$

Within each imbalanced training fold, the minority class was the control class. For each synthetic subject, one minority subject was sampled uniformly at random, and a partner was selected uniformly from that subject’s five nearest minority neighbors under squared chordal distance. Synthetic subjects were generated until the two class counts were equal. The same sampled value of t was used for the Grassmann and Euclidean interpolations. For numerical stabilization, the implementation applied a QR factorization to $Q_{\text{syn}}(t)$. The resulting invertible right basis change preserves the represented Grassmann point. No synthetic subjects were generated for balanced UCI-489. Interpolating whole-subject representations along the manifold avoids creating synthetic recording rows and selects partners by subspace geometry, but it still interpolates high-dimensional summaries and therefore remains susceptible to the noise-amplification risks of oversampling [47].

Proposition 2 (*manifold validity*). In exact arithmetic, the nonzero-angle columns of $Q_a U$ and W_{ab} are mutually orthonormal, while zero-angle directions remain in the cosine term. Consequently,

$$Q_{\text{syn}}(t)^T Q_{\text{syn}}(t) = V [\cos^2(t\Theta) + \sin^2(t\Theta)] V^T = I_r. \quad (13)$$

Therefore, $Q_{\text{syn}}(t) \in \text{St}(r, d)$, the Stiefel manifold of orthonormal $d \times r$ frames; it represents a valid Grassmann point. Its endpoints are equivalent to $[Q_a]$ and $[Q_b]$. $\square$

2.8. Kernel Support Vector Machine, Metrics, and Complexity

For support vector machine (SVM) training [62], labels are recoded as $y_i \in \{-1, +1\}$, $Y = \text{diag}(y_1, \dots, y_n)$, and balanced class weighting sets $C_{y_i} = Cn / (2n_{y_i})$, where n_{y_i} is the number of training subjects in class y_i ; this is the balanced heuristic implemented in standard open-source libraries [63]. When the training classes are exactly balanced, as on UCI-489 and after G-SMOTE parity on PD-252, these weights reduce to $C_{y_i} = C$; the weighting is therefore active only for the unaugmented class-weighted comparators on imbalanced PD-252. The class-weighted dual and the decision function with bias b are

$$\begin{aligned} \max_{\alpha} \quad & 1^T \alpha - \frac{1}{2} \alpha^T Y K Y \alpha, \\ \text{subject to} \quad & 0 \leq \alpha_i \leq C_{y_i}, y^T \alpha = 0, \\ f(Z) \quad & = \sum_{i=1}^n \alpha_i y_i K(Z_i, Z) + b. \end{aligned} \quad (14)$$

Balanced accuracy (BA) [64] and macro-F1 [65] weight diagnostic classes symmetrically:

$$\begin{aligned} \text{BA} &= \frac{1}{2} (\text{TPR} + \text{TNR}), \\ \text{macro-F1} &= \frac{1}{2} (F1_{\text{PD}} + F1_{\text{Ctrl}}), \end{aligned} \quad (15)$$

where TPR is sensitivity, TNR is specificity, and $F1_{\text{PD}}$ and $F1_{\text{Ctrl}}$ are the class-wise F1 scores. The Matthews correlation coefficient is a common alternative [66]; balanced accuracy and macro-F1 were retained because both weight the two classes symmetrically and are directly interpretable as class-wise rates. For $F = 5$ outer folds, a metric m is summarized by

$$\begin{aligned} \bar{m} &= \frac{1}{F} \sum_{k=1}^F m_k, \\ \text{SE}(\bar{m}) &= \left[\frac{1}{F(F-1)} \sum_{k=1}^F (m_k - \bar{m})^2 \right]^{1/2}. \end{aligned} \quad (16)$$

The reported standard error (SE) is a descriptive measure of fold-to-fold variability and is not an inferential confidence interval.

For n subjects and rank $r = 3$, subject SVDs require $O(ndr^2)$, the Grassmann Gram matrix requires $O(n^2 dr^2)$, and the Euclidean block requires $O(n^2 d)$. These bounds exclude the kernel-SVM optimization and the additional G-SMOTE neighbor search, which can dominate as n grows; all components remain tractable for the present cohort sizes $n = 80$ and $n = 252$.

2.9. Formal Effect of Record-Wise Leakage

The mechanism illustrated in Figure 2 can be stated formally, following record-wise versus subject-wise analyses in clinical machine learning [18,20,21]. Let (X, y) be a random validation recording and its label, let A be the event that the recording belongs to a person represented in training, and let h be a record-level classifier. With A^c the complement of A , define

$$\begin{aligned} R_{\text{seen}} &= E[\ell(h(X), y) | A], \\ R_{\text{new}} &= E[\ell(h(X), y) | A^c]. \end{aligned} \quad (17)$$

A record-wise split then estimates

$$\begin{aligned} R_{\text{row}} &= \Pr(A) R_{\text{seen}} + [1 - \Pr(A)] R_{\text{new}}, \\ R_{\text{row}} - R_{\text{new}} &= \Pr(A) (R_{\text{seen}} - R_{\text{new}}). \end{aligned} \quad (18)$$

If repeated recordings from previously observed people are easier to classify than recordings from unseen people, then $R_{\text{seen}} < R_{\text{new}}$, and the record-wise estimate is optimistically biased in direct proportion to $\Pr(A)$ (Figure 2C). Subject-disjoint folds enforce $\Pr(A) = 0$ by construction.

2.10. Experimental Protocol

Outer folds were generated by five-fold stratified subject-level cross-validation with shuffling and a random seed of 42 (Figure 5A). For each outer fold, the recording-level standardizer was fitted only to recordings from training subjects. Each kernel coefficient was estimated from the positive squared pairwise distances δ^2 of the corresponding view over the training subjects, after G-SMOTE augmentation on PD-252, as $\gamma = [2 \text{ median}(\delta^2) + 10^{-12}]^{-1}$. Rank $r = 3$, fusion weight $w = 0.5$, and SVM penalty $C = 1$ were fixed across datasets; no per-dataset hyperparameter search was performed, thereby avoiding reuse of the outer test folds for hyperparameter selection [23,24]. The same protocol was applied to both cohorts. Within each outer training fold, resampling was governed solely by the observed training-fold class counts: if the counts were unequal, the minority class was raised to the majority count using G-SMOTE for the kernel models and ordinary SMOTE for the Euclidean comparators; if the counts were equal, resampling was the identity operation and no synthetic subjects were added. All UCI-489 training folds were balanced, so no synthetic subjects were generated there, whereas every PD-252 training fold activated the branch. The comparator set was identical across cohorts: the fused model; a matched Euclidean RBF SVM with fusion weight $w = 0$, which shares the summary features, the resampling, and the kernel family of the fused model and differs from it only in the Grassmann kernel term (Figure 5B); a Grassmann-only kernel SVM with $w = 1$; class-weighted logistic regression; and class-weighted linear SVM. On PD-252, where resampling is active, SMOTE plus logistic regression and SMOTE plus Euclidean RBF SVM were added; on UCI-489 these coincide with their unresampled counterparts. All methods were implemented in Python, and the implementation is available in the project repository (see the Data Availability Statement). Table 3 summarizes all fixed protocol choices. All comparators were implemented in the released codebase and verified against their stated configurations; the term audited comparators refers to this set, as distinct from values reported by other studies.

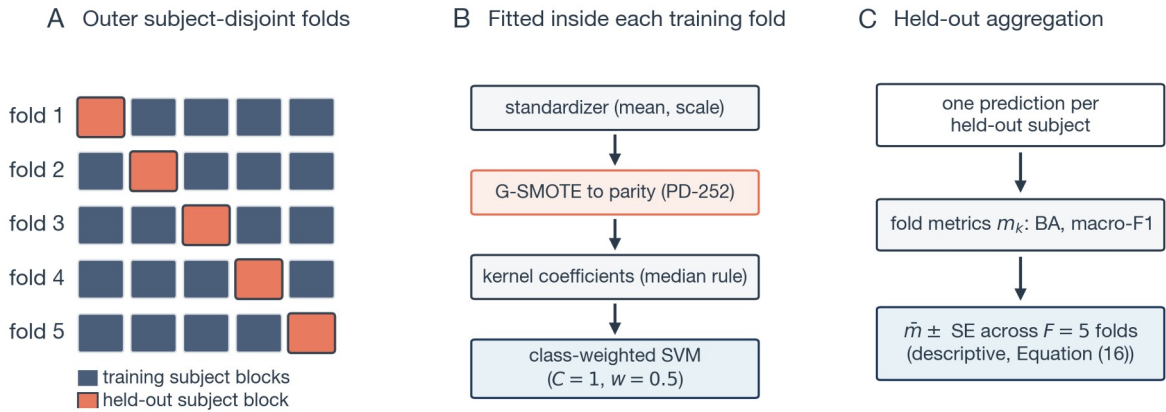
Each comparison isolates one design decision. Class-weighted logistic regression and class-weighted linear SVM probe whether simple decision boundaries on the Euclidean summary already suffice; SMOTE plus logistic regression tests a standard imbalanced-learning remedy [44]. The matched Euclidean RBF SVM is the critical ablation for the kernel term: any advantage of the fused model over it is attributable to the geometric kernel component, conditional on the shared augmentation, which itself selects partners by chordal distance (Section 2.7). SMOTE plus Euclidean RBF SVM is the fully Euclidean kernel pipeline; comparing it with the matched ablation isolates the geodesic augmentation, and comparing it with the fused model measures the complete geometric pipeline. On PD-252, G-SMOTE randomly samples anchor subjects and neighbors and draws interpolation coefficients, so performance varies with the augmentation seed. To isolate this source of variability, the fixed five-fold evaluation was repeated for 24 distinct augmentation seeds while the outer-fold assignments, preprocessing, hyperparameters, and all non-augmentation settings were held fixed. Within each seed and fold, the fused model and the matched Euclidean RBF ablation used the same augmented training set, so their difference is paired at the seed level. All PD-252 headline results use the reference implementation’s default augmentation seed, which is archived and regenerable; the seed analysis of Section 3.3 characterizes variability around it.

Table 3. Fixed experimental protocol. All learned components are fitted inside the outer training folds only.

Component	Setting
Outer evaluation	five-fold stratified subject-level cross-validation, shuffled, seed 42
Standardization	per-coordinate mean/scale from training recordings; zero-variance coordinates assigned unit scale
Subject representation	rank $r = 3$ thin SVD (Grassmann view) + mean/dispersion vector (Euclidean view)
Kernel coefficients	$\gamma = [2 \text{ median}(\delta^2) + 10^{-12}]^{-1}$ from positive squared pairwise training distances (augmented training set on PD-252)

Fusion weight / penalty	$w = 0.5$; class-weighted SVM with $C = 1$; no per-dataset tuning
Resampling rule	training-fold class-count gate: identity when balanced (all UCI-489 folds); G-SMOTE to class parity when imbalanced (all PD-252 folds), five nearest minority neighbors under chordal distance
Comparators	fused, matched Euclidean RBF, and Grassmann-only kernel SVMs (fusion weights 0.5, 0, and 1); class-weighted logistic regression; class-weighted linear SVM; SMOTE + logistic regression and SMOTE + Euclidean RBF where resampling is active
Metrics	balanced accuracy and macro-F1, mean $\pm$ fold-level SE; pooled out-of-fold confusion counts
Sensitivity analysis	20 outer subject partitions; rank $r = 2$; 24 augmentation seeds; held-out permutation of Euclidean summary blocks (20 draws per family and fold), no refitting

292



293

294 **Figure 5.** Evaluation protocol. (A) Five-fold subject-disjoint partition; in each outer fold one subject block (orange)
295 is held out and never influences any fitted component. (B) Everything that learns from data—the standardizer,
296 conditional G-SMOTE (activated only in imbalanced training folds), the kernel coefficients, and the SVM—is fitted
297 inside the training fold only. (C) Exactly one prediction is issued per held-out subject, fold metrics are computed at
298 the subject level, and Equation (16) aggregates them descriptively.

299 2.11. Feature-Family Permutation Sensitivity

300 For feature family g , the fold-level decrease in performance and its summary are

$$\begin{aligned}
 I_{gk} &= m_k - m_k^{\pi_g}, \hat{I}_g = \frac{1}{F} \sum_{k=1}^F I_{gk}, \\
 \text{SE}(\hat{I}_g) &= \left[\frac{1}{F(F-1)} \sum_{k=1}^F (I_{gk} - \hat{I}_g)^2 \right]^{1/2}.
 \end{aligned}
 \tag{19}$$

302 Sensitivity of the Euclidean summary was evaluated on each held-out fold without refitting. The
303 coordinates of s_i associated with family g were jointly permuted across held-out subjects while Q_i and
304 all other summary coordinates were held fixed. In Equation (19), m_k is the original held-out macro-F1
305 and $m_k^{\pi_g}$ is macro-F1 after permutation. Twenty permutation draws were made per family and fold with
306 a fixed random seed, the decrease was averaged over draws, and the six families follow the descriptor
307 taxonomy of the source dataset (column assignments are archived with the code).

308 2.12. Robustness Analyses

309 Three further analyses were prespecified after the first review of the design. First, the entire five-fold
310 evaluation was repeated for 20 alternative stratified subject partitions (StratifiedKFold seeds 0–19) with
311 the default augmentation rule, and paired differences between models were summarized across
312 partitions. Second, the fused model was rerun with rank $r = 2$, for which every subject matrix has full

rank, to test sensitivity to the rank choice and to the two rank-deficient subjects noted in Section 2.3. Third, on PD-252 the augmentation family was varied across the same 24 seeds as Section 3.3, so that the fused model, the matched Euclidean RBF ablation with G-SMOTE, and the fully Euclidean SMOTE plus Euclidean RBF pipeline were compared with paired draws. Fourth, the fusion weight was varied over $w \in \{0, 0.1, \dots, 0.8\}$ in every seed and partition run, and a nested variant selected w inside each training fold by repeated (three times five-fold) inner cross-validation on balanced accuracy over the same grid, so that the held-out folds were never used for selection. Exploratory alternatives (dimension-reduced, rank-one, centered, and block-balanced Grassmann views; decision-level fusion; a weight ensemble; other SVM penalties) are archived with the code. All robustness analyses used the fixed hyperparameters of Table 3 unless stated and are archived with the reference implementation.

3. Results

3.1. Performance Across Datasets

Figure 6 and Table 4 report every comparator under the single fixed protocol. On UCI-489, the fused model reached a balanced accuracy of 0.850 (SE 0.042) and a macro-F1 of 0.847 (SE 0.043); the matched Euclidean RBF ablation reached 0.863 (SE 0.041) and 0.860 (SE 0.042), a difference of one subject in one fold, whereas the Grassmann-only kernel reached 0.600 (SE 0.047) and 0.574 (SE 0.050). Both kernel models exceeded class-weighted logistic regression (0.775 and 0.770) and class-weighted linear SVM (0.713 and 0.704). On PD-252, at the archived default augmentation seed, the fused model reached 0.731 (SE 0.032) and 0.749 (SE 0.031), the matched Euclidean RBF ablation 0.743 (SE 0.021) and 0.745 (SE 0.019), and the Grassmann-only kernel 0.636 (SE 0.038) and 0.647 (SE 0.051); class-weighted logistic regression, class-weighted linear SVM, SMOTE plus logistic regression, and SMOTE plus Euclidean RBF reached 0.730 and 0.736, 0.722 and 0.730, 0.730 and 0.735, and 0.728 and 0.714, respectively. Thus, under one fixed configuration, the fused model was within one subject per fold of the best comparator on UCI-489 and within fold-level standard error of the best comparators on PD-252, while the Grassmann view alone was clearly insufficient on both cohorts. These are descriptive cross-validation estimates at one outer partition; Sections 3.3 and 3.5 characterize their variability, and no inferential test of model superiority was performed.

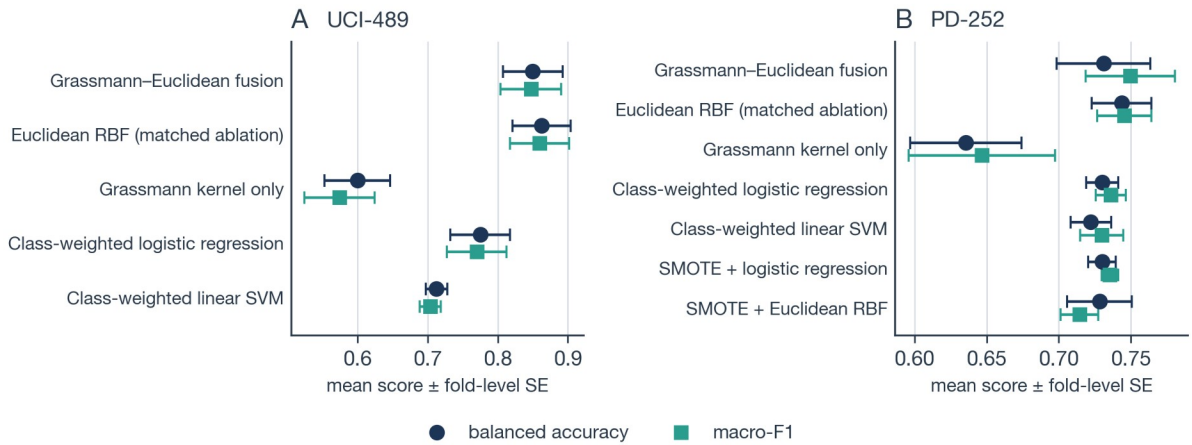


Figure 6. Subject-level scores of all comparators under the single fixed protocol for (A) class-balanced UCI-489 and (B) high-dimensional, class-imbalanced PD-252, drawn from the archived out-of-fold results (PD-252 at the archived default augmentation seed). Markers denote mean balanced accuracy (circles) and macro-F1 (squares); whiskers denote fold-level standard error across five outer folds. On UCI-489, the SMOTE-based comparators coincide with their unresampled counterparts because all training folds are balanced and are therefore not shown.

Table 4. Subject-level performance of every comparator under the single fixed protocol (mean $\pm$ fold-level standard error over five subject-disjoint folds; PD-252 at the archived default augmentation seed). SMOTE-based comparators are listed only where the resampling gate is active. Seed and partition variability is quantified in Tables 6 and 7.

Dataset	Model	Balanced acc.	Macro-F1
UCI-489	Grassmann–Euclidean fusion	0.850 $\pm$ 0.042	0.847 $\pm$ 0.043
UCI-489	Euclidean RBF (matched ablation)	0.863 $\pm$ 0.041	0.860 $\pm$ 0.042
UCI-489	Grassmann kernel only	0.600 $\pm$ 0.047	0.574 $\pm$ 0.050
UCI-489	Class-weighted logistic regression	0.775 $\pm$ 0.042	0.770 $\pm$ 0.043
UCI-489	Class-weighted linear SVM	0.713 $\pm$ 0.015	0.704 $\pm$ 0.015
PD-252	Grassmann–Euclidean fusion + G-SMOTE	0.731 $\pm$ 0.032	0.749 $\pm$ 0.031
PD-252	Euclidean RBF + G-SMOTE (matched ablation)	0.743 $\pm$ 0.021	0.745 $\pm$ 0.019
PD-252	Grassmann kernel only + G-SMOTE	0.636 $\pm$ 0.038	0.647 $\pm$ 0.051
PD-252	Class-weighted logistic regression	0.730 $\pm$ 0.011	0.736 $\pm$ 0.010
PD-252	Class-weighted linear SVM	0.722 $\pm$ 0.014	0.730 $\pm$ 0.015
PD-252	SMOTE + logistic regression	0.730 $\pm$ 0.009	0.735 $\pm$ 0.006
PD-252	SMOTE + Euclidean RBF	0.728 $\pm$ 0.022	0.714 $\pm$ 0.013

3.2. Operating Characteristics

The UCI-489 fused model produced 34 true-negative (control), six false-positive, six false-negative, and 34 true-positive (PD) classifications, yielding sensitivity and specificity of 0.850. On PD-252, under the archived default-seed G-SMOTE realization (Section 3.3), the pooled confusion matrix contained 175 true-positive (PD), 13 false-negative, 34 true-negative (control), and 30 false-positive classifications, corresponding to sensitivity 0.931, specificity 0.531, positive predictive value 0.854, and negative predictive value 0.723 (Figure 7, Table 5). Sensitivity and specificity are prevalence-free, whereas PPV and NPV depend on this cohort’s 188/64 composition and the default decision threshold and do not transport to lower-prevalence screening settings. Rates computed from pooled confusion matrices are descriptive and need not equal the arithmetic mean of fold-level metrics. At the default threshold, the modest PD-252 specificity implies a potentially substantial false-referral burden, especially in lower-prevalence screening populations. This operating profile warrants further evaluation of the method as a research-stage screening aid and does not support its use for autonomous diagnosis.

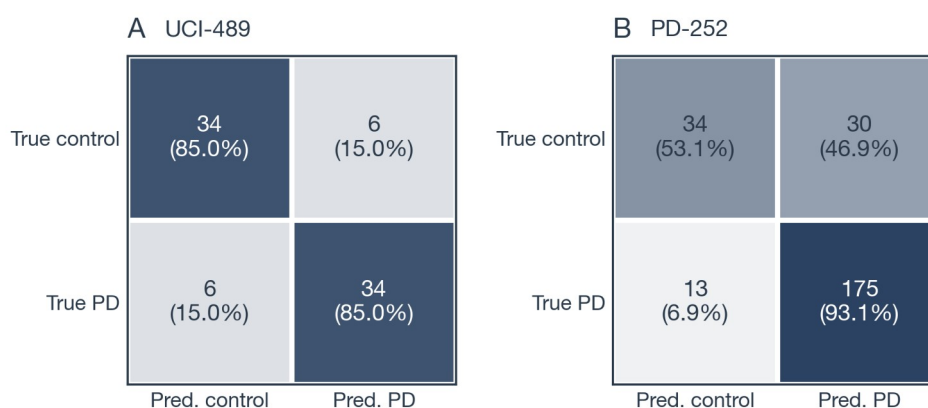


Figure 7. Pooled out-of-fold confusion matrices of the fused model for (A) UCI-489 and (B) PD-252, containing one prediction per subject; percentages are normalized within the true class. PD-252 counts are from the archived default-seed G-SMOTE realization (Section 3.3).

Table 5. Pooled out-of-fold operating characteristics computed from the confusion counts of Figure 7 (UCI-489: TP = 34, FN = 6, TN = 34, FP = 6; PD-252: TP = 175, FN = 13, TN = 34, FP = 30). TP, true positive; FN, false negative; TN, true negative; FP, false positive; PPV, positive predictive value; NPV, negative predictive value.

Dataset	Sensitivity	Specificity	PPV	NPV
UCI-489	0.850	0.850	0.850	0.850
PD-252	0.931	0.531	0.854	0.723

3.3. Sensitivity to the G-SMOTE Augmentation Seed

The PD-252 scores in Table 4 were obtained at the archived default augmentation seed. The paired analysis of Section 2.10 characterizes the sensitivity of both models and of their difference to the augmentation seed (Table 6). Across 24 seed runs, balanced accuracy was 0.739 ± 0.012 for the fused model and 0.731 ± 0.013 for the matched Euclidean ablation, and macro-F1 was 0.752 ± 0.013 and 0.729 ± 0.013 , respectively (mean $\pm$ standard deviation across seeds). The paired macro-F1 difference averaged $+0.023 \pm 0.015$ and was positive in 23 of 24 runs. The paired balanced-accuracy difference averaged $+0.009 \pm 0.014$, was positive in 16 of 24 runs, and reversed sign in 8 of 24. The archived default-seed realization of Tables 4 and 5 (0.731 and 0.749) lies within the seed ranges [0.722, 0.760] and [0.732, 0.777]. Across the same 24 seeds, the fully Euclidean SMOTE plus Euclidean RBF pipeline averaged 0.726 ± 0.009 and 0.718 ± 0.010 ; replacing ordinary SMOTE by G-SMOTE in the Euclidean RBF model changed balanced accuracy by $+0.005 \pm 0.016$ and macro-F1 by $+0.011 \pm 0.016$ (positive in 15 and 17 of 24 seeds), and the complete geometric pipeline exceeded the fully Euclidean pipeline by $+0.014 \pm 0.015$ and $+0.034 \pm 0.016$ (positive in 19 and 24 of 24 seeds; Table 7). Thus, conditional on the fixed outer folds, the macro-F1 margin was directionally consistent in all but one seed run, whereas the balanced-accuracy margin can be reversed by individual draws.

Table 6. Sensitivity of the PD-252 comparison to the G-SMOTE augmentation seed: paired evaluation of the fused model and the matched Euclidean RBF ablation across 24 distinct seeds with outer folds, preprocessing, and hyperparameters held fixed. Values are means $\pm$ standard deviations (SD) across seeds; Δ denotes the paired difference (fused minus ablation) computed within each seed, summarized by its mean $\pm$ SD, median [interquartile range, IQR], and sign count.

Metric	Fused	Ablation	Δ mean $\pm$ SD	Δ median [IQR]	$\Delta > 0$
Balanced accuracy	0.739 ± 0.012	0.731 ± 0.013	$+0.009 \pm 0.014$	$+0.008 [-0.002, +0.018]$	16 / 24
Macro-F1	0.752 ± 0.013	0.729 ± 0.013	$+0.023 \pm 0.015$	$+0.020 [+0.012, +0.034]$	23 / 24

Each seed-level value is the five-fold cross-validation point estimate under fixed outer folds; standard deviations describe variation across the 24 augmentation seeds and are distinct from the fold-level standard errors of Table 4. Δ ranges: balanced accuracy $[-0.013, +0.042]$; macro-F1 $[-0.001, +0.061]$. Per-seed values are archived in the project repository.

3.4. Feature-Family Sensitivity

The TQWT summary block produced the largest mean decrease in held-out macro-F1, at 0.149 with an SE of 0.020, followed by the MFCC summary block, at 0.046 with an SE of 0.008 (Figure 8); within a fold, the standard deviation across the 20 permutation draws averaged 0.067 for TQWT. The vocal /time-frequency, wavelet, IMF /EMD, and demographic families produced mean decreases near zero. These exploratory values measure dependence of the fitted model on the Euclidean summary coordinates under correlated features; they do not establish biological causality and, because only the Euclidean view is permuted, do not quantify the dependence of the full two-view model on each family. The prominence of the TQWT family is consistent with the descriptor engineering of the source dataset, where tunable Q-factor features were introduced precisely for Parkinsonian voice [31,55].

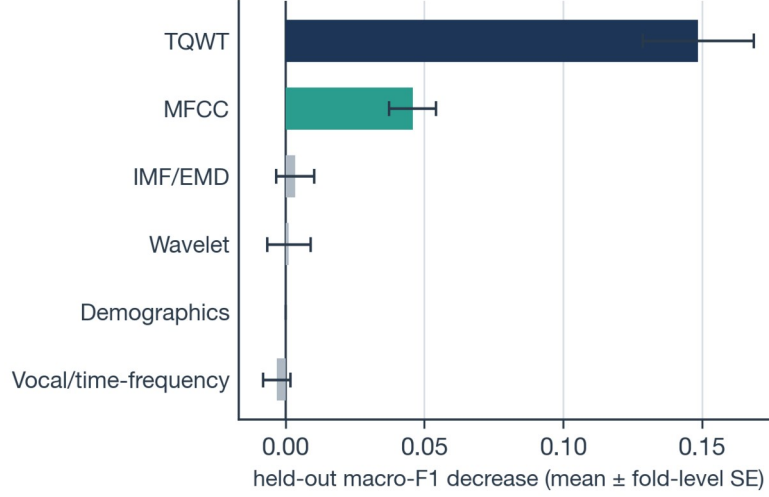


Figure 8. PD-252 feature-family permutation sensitivity of the fused model on the Euclidean summary view under the archived default-seed realization (20 permutation draws per family and fold). Positive values denote a decrease in held-out macro-F1 after permutation, and whiskers denote fold-level standard error. TQWT denotes tunable Q-factor wavelet transform; MFCC denotes Mel-frequency cepstral coefficients; IMF/EMD denotes intrinsic mode function/empirical mode decomposition; Demographics denotes the dataset-provided sex/gender covariate.

3.5. Robustness to the Outer Partition, the Rank, and the Augmentation Family

Figure 9 and Table 7 summarize the robustness analyses of Section 2.12. Across 20 alternative outer subject partitions, the fused model and the matched Euclidean RBF ablation were indistinguishable on both cohorts: the paired balanced-accuracy difference averaged $+0.004 \pm 0.013$ on UCI-489 (positive in 8 of 20 partitions, zero in 8) and -0.000 ± 0.016 on PD-252 (positive in 8 of 20), and the paired macro-F1 difference averaged $+0.005 \pm 0.013$ (11 of 20) and $+0.008 \pm 0.017$ (13 of 20). Against the fully Euclidean SMOTE plus Euclidean RBF pipeline, the fused model retained a macro-F1 advantage of $+0.028 \pm 0.019$ on PD-252 (18 of 20 partitions) with a balanced-accuracy difference of $+0.011 \pm 0.018$ (15 of 20); on UCI-489, where resampling is inactive, that comparison coincides with the ablation comparison. Across partitions the fused model exceeded class-weighted logistic regression on UCI-489 ($+0.028 \pm 0.021$ balanced accuracy, 16 of 20) but not on PD-252 (-0.004 ± 0.016 , 8 of 20). Reducing the rank to $r=2$ changed the fused model by less than one fold-level standard error (UCI-489: 0.838 and 0.834; PD-252: 0.739 and 0.755), so the two rank-deficient subjects do not drive the results. Partition-to-partition standard deviations (0.013–0.022) were of the same order as the augmentation-seed standard deviations of Table 6, and both are smaller than the fold-level standard errors of Table 4. Figure 10 shows the sensitivity of the comparison to the fusion weight. On PD-252, every weight from 0.1 to 0.7 exceeded the ablation in macro-F1 in at least 20 of 24 seeds, and weights from 0.1 to 0.5 exceeded it in balanced accuracy in 13 to 22 of 24 seeds, with the largest margins at $w=0.3$ ($+0.013$ and $+0.021$); across partitions the macro-F1 advantage persisted for $w \leq 0.7$ (14 of 20 at $w=0.3$), whereas balanced accuracy exceeded the ablation for $w \leq 0.3$ and fell below it for $w \geq 0.6$. On UCI-489 the optimum lay at larger weights ($w=0.7$: $+0.011$ in both metrics, 13 and 14 of 20 partitions). Because the optimal weight is cohort-dependent, the nested variant of Section 2.12 was also evaluated: it exceeded the ablation in balanced accuracy in 20 of 24 seeds and 13 of 20 partitions on PD-252 ($+0.005 \pm 0.007$ and $+0.001 \pm 0.010$) and in macro-F1 in 22 of 24 and 14 of 20 ($+0.011 \pm 0.008$ and $+0.004 \pm 0.010$), and was equal or better in 16 of 20 partitions on UCI-489 (Table 7). A tuning-free ensemble averaging the fused decision values over $w \in [0.1, 0.7]$ behaved like the fixed weight, and none of the archived exploratory variants exceeded the fixed-weight fused model robustly.

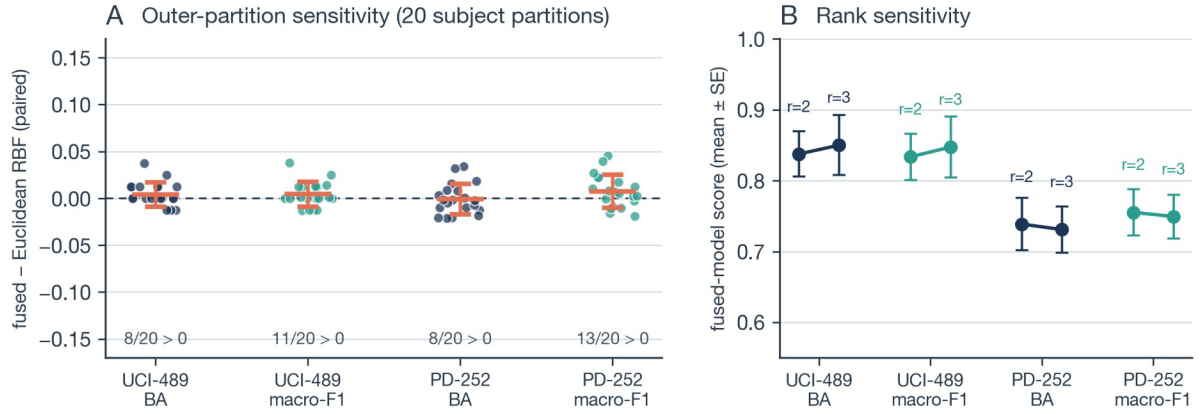


Figure 9. Robustness of the fused model under the fixed protocol. **(A)** Paired differences between the fused model and the matched Euclidean RBF ablation across 20 alternative outer subject partitions (points; coral bars denote mean $\pm$ SD; counts give the number of partitions with a positive difference), for balanced accuracy (BA, navy) and macro-F1 (teal) on both cohorts. **(B)** Fused-model scores at rank $r=2$ versus $r=3$ (mean $\pm$ fold-level SE) at the default partition.

Table 7. Robustness analyses under the fixed protocol (Section 2.12). Each row pairs two models within the same partition or the same augmentation seed (nested-weight fusion selects $\mathbf{w}$ by repeated inner cross-validation inside each training fold); values are means $\pm$ standard deviations across the 20 outer subject partitions or the 24 augmentation seeds, Δ is the paired difference (model A minus model B), and the last column counts strictly positive differences (ties noted where they occur). On UCI-489, SMOTE + Euclidean RBF coincides with the matched RBF ablation because resampling is inactive.

Analysis (A vs. B)	Cohort	Metric	Model A	Model B	Δ (A - B)	$\Delta > 0$
Outer partition (20): fused vs. matched RBF	UCI-489	BA	0.853 ± 0.013	0.848 ± 0.014	$+0.004 \pm 0.013$	8/20 (8 ties)
	UCI-489	macro-F1	0.851 ± 0.014	0.846 ± 0.015	$+0.005 \pm 0.013$	11/20 (5 ties)
	PD-252	BA	0.725 ± 0.014	0.725 ± 0.020	0.000 ± 0.016	8/20
	PD-252	macro-F1	0.737 ± 0.016	0.729 ± 0.022	$+0.008 \pm 0.017$	13/20
Outer partition (20): fused vs. SMOTE + RBF	PD-252	BA	0.725 ± 0.014	0.714 ± 0.018	$+0.011 \pm 0.018$	15/20
	PD-252	macro-F1	0.737 ± 0.016	0.709 ± 0.020	$+0.028 \pm 0.019$	18/20
Outer partition (20): fused vs. logistic regression	UCI-489	BA	0.853 ± 0.013	0.824 ± 0.024	$+0.028 \pm 0.021$	16/20 (3 ties)
	UCI-489	macro-F1	0.851 ± 0.014	0.822 ± 0.025	$+0.029 \pm 0.021$	18/20
	PD-252	BA	0.725 ± 0.014	0.729 ± 0.017	-0.004 ± 0.016	8/20
	PD-252	macro-F1	0.737 ± 0.016	0.734 ± 0.017	$+0.003 \pm 0.017$	13/20
Augmentation seed (24): fused vs. SMOTE + RBF	PD-252	BA	0.739 ± 0.012	0.726 ± 0.009	$+0.014 \pm 0.015$	19/24
	PD-252	macro-F1	0.752 ± 0.013	0.718 ± 0.010	$+0.034 \pm 0.016$	24/24
Augmentation seed (24): G-SMOTE vs. SMOTE (RBF model)	PD-252	BA	0.731 ± 0.013	0.726 ± 0.009	$+0.005 \pm 0.016$	15/24
	PD-252	macro-F1	0.729 ± 0.013	0.718 ± 0.010	$+0.011 \pm 0.016$	17/24
Augmentation seed (24): nested-weight fusion vs. matched	PD-252	BA	0.736 ± 0.010	0.731 ± 0.013	$+0.005 \pm 0.007$	20/24

RBF						
	PD-252	macro-F1	0.740 ± 0.012	0.729 ± 0.013	$+0.011 \pm 0.008$	22/24
Outer partition (20):	UCI-489	BA	0.851 ± 0.015	0.848 ± 0.014	$+0.003 \pm 0.013$	7/20 (9 ties)
nested-weight fusion vs. matched RBF						
	UCI-489	macro-F1	0.850 ± 0.016	0.846 ± 0.015	$+0.003 \pm 0.014$	8/20 (7 ties)
	PD-252	BA	0.726 ± 0.016	0.725 ± 0.020	$+0.001 \pm 0.010$	13/20 (1 ties)
	PD-252	macro-F1	0.733 ± 0.017	0.729 ± 0.022	$+0.004 \pm 0.010$	14/20 (1 ties)

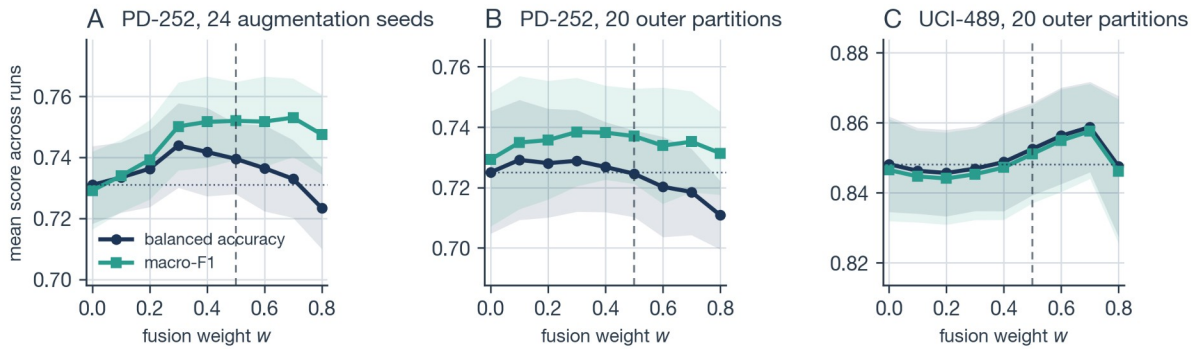


Figure 10. Sensitivity of the fused model to the fusion weight w under the fixed protocol: mean balanced accuracy (navy circles) and macro-F1 (teal squares) across (A) the 24 augmentation seeds on PD-252, (B) the 20 outer partitions on PD-252, and (C) the 20 outer partitions on UCI-489. Shaded bands denote $\pm$ one standard deviation across runs, the dashed line marks the prespecified weight $w = 0.5$, and the dotted line marks the balanced accuracy of the matched Euclidean RBF ablation ($w = 0$).

4. Discussion

The central methodological contribution is the alignment of the evaluation unit with the clinical unit. Subject-disjoint validation asks whether a model generalizes to an unseen person. The Grassmann representation uses the dependence among repeated recordings rather than treating the recordings as independent replicates. Basis invariance removes arbitrary coordinate choices, while the Euclidean view restores the coordinate-level location and dispersion magnitudes that a pure subspace representation discards.

Under the prespecified configuration, the fused model improved macro-F1 over the matched Euclidean RBF ablation in 23 of 24 augmentation seeds and 13 of 20 outer partitions on PD-252 and was equal or better in 16 of 20 partitions on UCI-489, whereas its balanced-accuracy advantage at the fixed weight was confined to the seed study (16 of 24) and vanished across partitions. The weight sweep explains this pattern: the fused model exceeds the ablation over a broad range of weights on both cohorts, but the optimal weight is cohort-dependent (about 0.3 on PD-252 and 0.7 on UCI-489), so a single fixed weight is a compromise between the two regimes. Selecting the weight inside each training fold made the balanced-accuracy improvement consistent (20 of 24 seeds and 13 of 20 partitions) at a smaller magnitude. The clearest benefit of the geometric machinery appeared when its two components were combined: the complete pipeline exceeded the fully Euclidean SMOTE plus Euclidean RBF pipeline in macro-F1 in 24 of 24 seeds and 18 of 20 partitions on PD-252, with the geodesic augmentation contributing about a third of that margin. None of these counts is population-level inference about model superiority. With respect to the objectives of Section 1, the first was addressed analytically, without a matched record-wise experiment: Equations (17) and (18) formalize the difference between record-wise and unseen-person risks. The second receives an affirmative but modest answer: adding

the Grassmann kernel to a matched nonlinear Euclidean model improved subject-level macro-F1 consistently, improved balanced accuracy when the weight was selected within the training data, and, as part of the complete geometric pipeline, improved macro-F1 over a fully Euclidean kernel pipeline on the imbalanced cohort. Whenever repeated recordings from previously observed people are easier to classify than recordings from unseen people, record-wise scores are optimistically biased in direct proportion to the probability that a validation recording belongs to an already-observed person. Subject-disjoint folds remove that term entirely. Reported subject-level scores are therefore expected to be lower than record-wise figures reported on the same corpora, but they answer the question that a screening deployment poses [20,22].

An evaluation-focused perspective also clarifies how these results should be interpreted in relation to the wider literature. Reported accuracies for voice-based PD detection vary widely across studies, and validation strategy is a recognized source of that variation [16,17]. Record-wise splitting remains common in secondary analyses of repeated-recording corpora, whereas replication-aware methods have also been developed for UCI-489 [29]. Balanced accuracy and macro-F1 were used because both cohorts were evaluated at the subject level and PD-252 had unequal class sizes. Both metrics weight the diagnostic classes symmetrically rather than rewarding majority-class predictions [64,65]. Fold-level standard errors are reported descriptively, and no significance test is claimed: five outer folds provide a descriptive summary of variability, not a basis for inferential comparison [25].

A separate research-stage web interface (Figure 11) demonstrates browser-based capture with a commodity microphone for three speech tasks (sustained vowel, read passage using an excerpt of the Rainbow Passage [67], and free speech), together with an illustrative longitudinal display. The application was not connected to either evaluated cohort and was not assessed for usability, calibration, or clinical performance. Figure 12 places the prototype in a proposed, rather than validated, deployment workflow and distinguishes the components evaluated here from capture and decision-support stages that were not; the evaluated models used only three repeated sustained-vowel recordings per subject, and constructing a common representation from heterogeneous application tasks remains future work. Such low-friction, repeated measurement is consistent with technology roadmaps for PD care [50,51]. The application is a research-stage prototype, not an autonomous diagnostic system or medical device, and no application-collected data were used in the analyses reported here. Prospective use requires explicit privacy, consent, security, accessibility conformance, calibration, and clinical governance procedures.

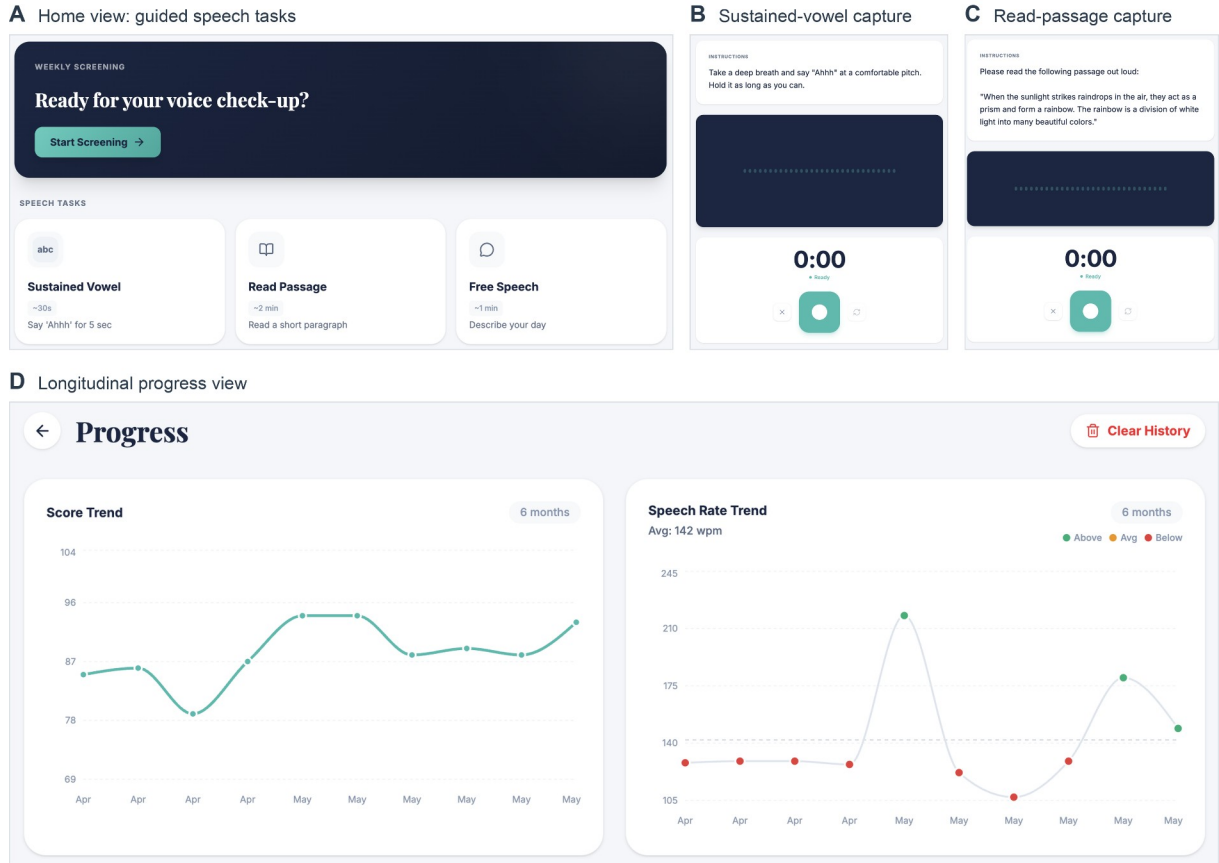


Figure 11. Deployed VoxNeuro web application (research-stage prototype). (A) Home view listing the three guided speech tasks. (B) Sustained-vowel capture screen. (C) Read-passages capture screen showing an excerpt from the Rainbow Passage. (D) Longitudinal progress view with illustrative entries. The application requires only a commodity microphone and a web browser; no application-collected data were used in the analyses reported here.

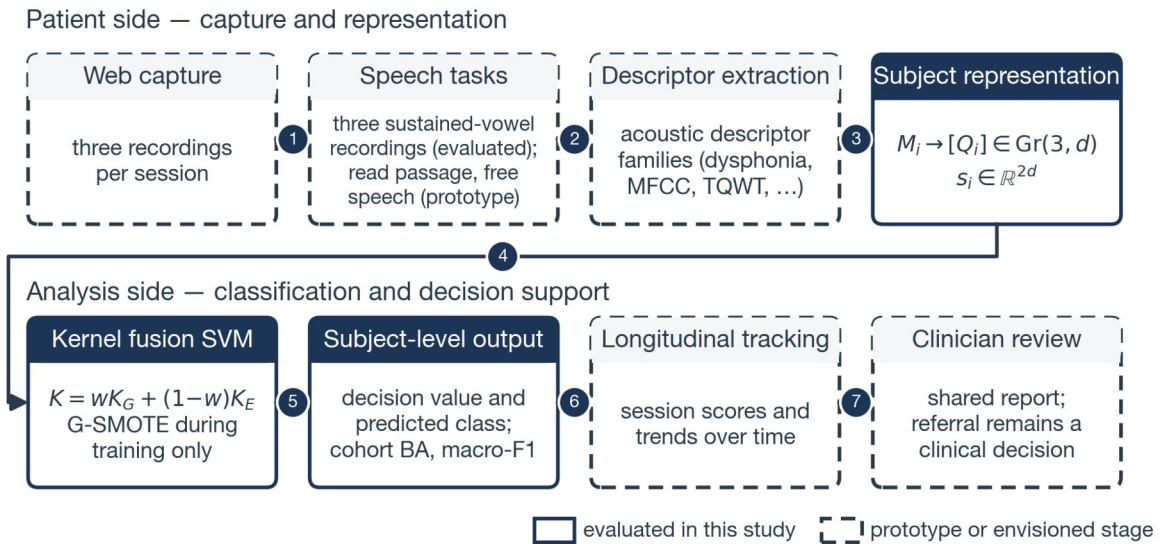


Figure 12. Study pipeline and envisioned deployment workflow of the VoxNeuro prototype. Solid boxes mark the components evaluated in this study on the two public cohorts (subject representation, kernel fusion with G-SMOTE, and subject-level scoring); dashed boxes mark capture and decision-support stages of the research-stage prototype that were not evaluated here. Descriptor extraction in this study used the dataset-provided engineered features, and any referral decision remains a clinical judgment. The evaluated models used only three repeated sustained-vowel recordings per subject; the read-passage and free-speech tasks are prototype capture tasks that were not used in any evaluated model.

This study has limitations. The speech cohorts contain retrospective engineered features rather than prospectively acquired raw audio. The study included no probability calibration [68,69], decision-curve analysis [70], independent clinical test cohort, device-shift evaluation, language-shift audit, or prespecified subgroup analysis. Sensitivity to the fusion weight, the outer partition, the augmentation seed, the augmentation family, and the subspace rank was characterized, but sensitivity to the SVM penalty, neighbor count, and median rule for the kernel coefficients was not, and the gains attributable to the geometric kernel term are small relative to fold-level variability. The comparator set did not include deep models, and pretrained audio models could not be evaluated because raw audio was unavailable. The Grassmann view alone was clearly insufficient in both cohorts (balanced accuracy 0.600 and 0.636). Because high PD-252 sensitivity coexists with modest specificity, a credible system should support referral and longitudinal review rather than replace neurological assessment. Future work should preregister preprocessing and thresholds and report in accordance with structured guidance for clinical prediction models [71,72]; collect repeated raw speech prospectively; and evaluate calibration, test–retest reliability, fairness, and clinician-facing decision consequences. The two-view construction extends naturally to additional recording tasks and views through further positive-semidefinite kernel terms, with weights selected by nested multiple-kernel optimization [42,43].

Reproducibility safeguards included fixed subject-disjoint folds, training-fold-only preprocessing, fixed hyperparameters, saved fold-level metrics, pooled confusion counts, and restriction of reported results to audited comparator implementations. The project repository archives, under a pinned dependency environment, subject-level out-of-fold predictions with fold identifiers for both cohorts, the unified comparator results, the 24-seed paired metrics, the partition, rank, augmentation-family, and permutation outputs with the feature-family column map, and the scripts that generate them, so that every reported aggregate in Tables 4–7 and Figures 6–10 can be regenerated exactly.

5. Conclusions

VoxNeuro provides a leakage-safe framework for repeated Parkinsonian speech. Each subject is represented by a Grassmann subspace and a Euclidean mean–dispersion vector. The corresponding positive-semidefinite kernels are fused, and minority subjects are synthesized along manifold geodesics only within training folds. Under one fixed configuration, the fused model reached balanced accuracies of 0.850 on UCI-489 and 0.731 on PD-252, improved macro-F1 over a matched Euclidean RBF ablation in 23 of 24 augmentation seeds and 13 of 20 outer partitions on PD-252, and, with the fusion weight selected within training folds, improved balanced accuracy in 20 of 24 seeds; the complete geometric pipeline exceeded a fully Euclidean kernel pipeline in macro-F1 in 24 of 24 seeds and 18 of 20 partitions. The deployed web application implements guided capture and longitudinal visualization as a research-stage prototype. Clinically credible digital biomarkers require auditable implementations, transparent uncertainty, and evaluation designs that test generalization to unseen people; prospective external validation is the necessary next step before any clinical use.

Author Contributions: Conceptualization, L.K.; methodology, L.K.; software, L.K.; validation, L.K. and M.W.; formal analysis, L.K.; investigation, L.K. and M.W.; data curation, L.K.; writing—original draft preparation, L.K.; writing—review and editing, L.K. and M.W.; visualization, L.K.; project administration, L.K. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Ethical review and approval were not required for this study, as it consisted exclusively of a secondary computational analysis of publicly available, fully de-identified datasets; no participants were recruited, and no new human data were collected.

Informed Consent Statement: Not applicable. The study analyzed publicly available, de-identified datasets and did not involve direct human participation; informed consent and ethics procedures for the original data collections are documented in the source publications cited in Section 2.1.

Data Availability Statement: The datasets analyzed in this study are openly available from the UCI Machine Learning Repository: the Parkinson Dataset with Replicated Acoustic Features (UCI-489; UCI dataset 489, doi:10.24432/C5701F) and the Parkinson’s Disease Classification dataset (PD-252; UCI dataset 470,

doi:10.24432/C5MS4X). Source code is available at <https://github.com/khan-laiba/voxneuro> (release v1.2.0); the repository also archives, under the pinned dependency environment, subject-level out-of-fold prediction files for both cohorts, the unified comparator results, the per-seed outputs of the PD-252 seed-sensitivity and augmentation-family analyses, the outer-partition and rank sensitivity results, the fusion-weight sweep, the nested-selection and exploratory-variant outputs, and the permutation analysis with its feature-family column map. The deployed research-stage web application is available at <https://voxneuro.org>. No new primary data were collected in this study.

Acknowledgments: During the preparation of this manuscript, the authors used Claude (Anthropic; Claude Opus and Claude Fable models, 2026) to draft and edit manuscript text, format the manuscript, write the code that draws the schematic Figures 1–5 and 12 and the data plots of Figures 6–10 from the archived results, and assemble the screenshot panels of Figure 11, and used GPT-5.6 Sol (OpenAI; Codex CLI 0.151.0) to review manuscript language and figure design. These tools were not used to generate or alter study data or reported numerical results. The authors have reviewed and edited all outputs and take full responsibility for the content of this publication.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Bloem, B.R.; Okun, M.S.; Klein, C. Parkinson's disease. *Lancet* **2021**, *397*, 2284–2303.
2. Kalia, L.V.; Lang, A.E. Parkinson's disease. *Lancet* **2015**, *386*, 896–912.
3. Poewe, W.; Seppi, K.; Tanner, C.M.; Halliday, G.M.; Brundin, P.; Volkmann, J.; Schrag, A.-E.; Lang, A.E. Parkinson disease. *Nat. Rev. Dis. Primers* **2017**, *3*, 17013.
4. Postuma, R.B.; Berg, D.; Stern, M.; Poewe, W.; Olanow, C.W.; Oertel, W.; Obeso, J.; Marek, K.; Litvan, I.; Lang, A.E.; et al. MDS clinical diagnostic criteria for Parkinson's disease. *Mov. Disord.* **2015**, *30*, 1591–1601.
5. Armstrong, M.J.; Okun, M.S. Diagnosis and treatment of Parkinson disease: A review. *JAMA* **2020**, *323*, 548–560.
6. GBD 2016 Parkinson's Disease Collaborators. Global, regional, and national burden of Parkinson's disease, 1990–2016: A systematic analysis for the Global Burden of Disease Study 2016. *Lancet Neurol.* **2018**, *17*, 939–953.
7. Dorsey, E.R.; Bloem, B.R. The Parkinson pandemic—A call to action. *JAMA Neurol.* **2018**, *75*, 9–10.
8. Logemann, J.A.; Fisher, H.B.; Boshes, B.; Blonsky, E.R. Frequency and cooccurrence of vocal tract dysfunctions in the speech of a large sample of Parkinson patients. *J. Speech Hear. Disord.* **1978**, *43*, 47–57.
9. Ho, A.K.; Iannsek, R.; Marigliani, C.; Bradshaw, J.L.; Gates, S. Speech impairment in a large sample of patients with Parkinson's disease. *Behav. Neurol.* **1998**, *11*, 131–137.
10. Rusz, J.; Cmejla, R.; Ruzickova, H.; Ruzicka, E. Quantitative acoustic measurements for characterization of speech and voice disorders in early untreated Parkinson's disease. *J. Acoust. Soc. Am.* **2011**, *129*, 350–367.
11. Little, M.A.; McSharry, P.E.; Roberts, S.J.; Costello, D.A.E.; Moroz, I.M. Exploiting nonlinear recurrence and fractal scaling properties for voice disorder detection. *Biomed. Eng. Online* **2007**, *6*, 23.
12. Little, M.A.; McSharry, P.E.; Hunter, E.J.; Spielman, J.; Ramig, L.O. Suitability of dysphonia measurements for telemonitoring of Parkinson's disease. *IEEE Trans. Biomed. Eng.* **2009**, *56*, 1015–1022.
13. Tsanas, A.; Little, M.A.; McSharry, P.E.; Ramig, L.O. Accurate telemonitoring of Parkinson's disease progression by noninvasive speech tests. *IEEE Trans. Biomed. Eng.* **2010**, *57*, 884–893.
14. Tsanas, A.; Little, M.A.; McSharry, P.E.; Spielman, J.; Ramig, L.O. Novel speech signal processing algorithms for high-accuracy classification of Parkinson's disease. *IEEE Trans. Biomed. Eng.* **2012**, *59*, 1264–1271.
15. Hlavnička, J.; Čmejla, R.; Tykalová, T.; Šonka, K.; Růžicka, E.; Rusz, J. Automated analysis of connected speech reveals early biomarkers of Parkinson's disease in patients with rapid eye movement sleep behaviour disorder. *Sci. Rep.* **2017**, *7*, 12.
16. Sedigh Malekroodi, H.; Lee, B.-i.; Yi, M. Voice-based detection of Parkinson's disease using machine and deep learning approaches: A systematic review. *Bioengineering* **2025**, *12*, 1279.
17. Moro-Velazquez, L.; Gomez-Garcia, J.A.; Arias-Londoño, J.D.; Dehak, N.; Godino-Llorente, J.I. Advances in Parkinson's disease detection and assessment using voice and speech: A review of the state-of-the-art. *Biomed. Signal Process. Control* **2021**, *66*, 102418.
18. Kaufman, S.; Rosset, S.; Perlich, C.; Stitelman, O. Leakage in data mining: Formulation, detection, and avoidance. *ACM Trans. Knowl. Discov. Data* **2012**, *6*, 15.
19. Kapoor, S.; Narayanan, A. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns* **2023**, *4*, 100804.
20. Saeb, S.; Lonini, L.; Jayaraman, A.; Mohr, D.C.; Kording, K.P. The need to approximate the use-case in clinical machine learning. *Gigascience* **2017**, *6*, gix019.

21. Little, M.A.; Varoquaux, G.; Saeb, S.; Lonini, L.; Jayaraman, A.; Mohr, D.C.; Kording, K.P. Using and understanding cross-validation strategies. Perspectives on Saeb et al. *Gigascience* **2017**, *6*, gix020.
22. Chaibub Neto, E.; Pratap, A.; Perumal, T.M.; Tummalacherla, M.; Snyder, P.; Bot, B.M.; Trister, A.D.; Friend, S.H.; Mangravite, L.; Omberg, L. Detecting the impact of subject characteristics on machine learning-based diagnostic applications. *NPJ Digit. Med.* **2019**, *2*, 99.
23. Cawley, G.C.; Talbot, N.L.C. On over-fitting in model selection and subsequent selection bias in performance evaluation. *J. Mach. Learn. Res.* **2010**, *11*, 2079–2107.
24. Varma, S.; Simon, R. Bias in error estimation when using cross-validation for model selection. *BMC Bioinformatics* **2006**, *7*, 91.
25. Varoquaux, G. Cross-validation failure: Small sample sizes lead to large error bars. *NeuroImage* **2018**, *180*, 68–77.
26. Arlot, S.; Celisse, A. A survey of cross-validation procedures for model selection. *Stat. Surv.* **2010**, *4*, 40–79.
27. Hastie, T.; Tibshirani, R.; Friedman, J. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, 2nd ed.; Springer: New York, NY, USA, 2009.
28. Pérez, C.J. *Parkinson Dataset with Replicated Acoustic Features*; UCI Machine Learning Repository, 2016. doi:10.24432/C5701F.
29. Naranjo, L.; Pérez, C.J.; Martín, J.; Campos-Roca, Y. A two-stage variable selection and classification approach for Parkinson's disease detection by using voice recording replications. *Comput. Methods Programs Biomed.* **2017**, *142*, 147–156.
30. Sakar, B.E.; Isenkul, M.E.; Sakar, C.O.; Sertbas, A.; Gurgun, F.; Delil, S.; Apaydin, H.; Kursun, O. Collection and analysis of a Parkinson speech dataset with multiple types of sound recordings. *IEEE J. Biomed. Health Inform.* **2013**, *17*, 828–834.
31. Sakar, C.O.; Serbes, G.; Gunduz, A.; Tunc, H.C.; Nizam, H.; Sakar, B.E.; Tutuncu, M.; Aydin, T.; Isenkul, M.E.; Apaydin, H. A comparative analysis of speech signal processing algorithms for Parkinson's disease classification and the use of the tunable Q-factor wavelet transform. *Appl. Soft Comput.* **2019**, *74*, 255–263.
32. Wong, Y.-C. Differential geometry of Grassmann manifolds. *Proc. Natl. Acad. Sci. USA* **1967**, *57*, 589–594.
33. Edelman, A.; Arias, T.A.; Smith, S.T. The geometry of algorithms with orthogonality constraints. *SIAM J. Matrix Anal. Appl.* **1998**, *20*, 303–353.
34. Absil, P.-A.; Mahony, R.; Sepulchre, R. *Optimization Algorithms on Matrix Manifolds*; Princeton University Press: Princeton, NJ, USA, 2008.
35. Hamm, J.; Lee, D.D. Grassmann discriminant analysis: A unifying view on subspace-based learning. In Proceedings of the 25th International Conference on Machine Learning, Helsinki, Finland, 5–9 July 2008; pp. 376–383.
36. Turaga, P.; Veeraraghavan, A.; Srivastava, A.; Chellappa, R. Statistical computations on Grassmann and Stiefel manifolds for image and video-based recognition. *IEEE Trans. Pattern Anal. Mach. Intell.* **2011**, *33*, 2273–2286.
37. Jayasumana, S.; Hartley, R.; Salzmann, M.; Li, H.; Harandi, M. Kernel methods on Riemannian manifolds with Gaussian RBF kernels. *IEEE Trans. Pattern Anal. Mach. Intell.* **2015**, *37*, 2464–2477.
38. Aronszajn, N. Theory of reproducing kernels. *Trans. Am. Math. Soc.* **1950**, *68*, 337–404.
39. Schölkopf, B.; Smola, A.J. *Learning with Kernels: Support Vector Machines, Regularization, Optimization, and Beyond*; MIT Press: Cambridge, MA, USA, 2002.
40. Shawe-Taylor, J.; Cristianini, N. *Kernel Methods for Pattern Analysis*; Cambridge University Press: Cambridge, UK, 2004.
41. Lanckriet, G.R.G.; Cristianini, N.; Bartlett, P.; El Ghaoui, L.; Jordan, M.I. Learning the kernel matrix with semidefinite programming. *J. Mach. Learn. Res.* **2004**, *5*, 27–72.
42. Gönen, M.; Alpaydin, E. Multiple kernel learning algorithms. *J. Mach. Learn. Res.* **2011**, *12*, 2211–2268.
43. Rakotomamonjy, A.; Bach, F.R.; Canu, S.; Grandvalet, Y. SimpleMKL. *J. Mach. Learn. Res.* **2008**, *9*, 2491–2521.
44. Chawla, N.V.; Bowyer, K.W.; Hall, L.O.; Kegelmeyer, W.P. SMOTE: Synthetic minority over-sampling technique. *J. Artif. Intell. Res.* **2002**, *16*, 321–357.
45. He, H.; Garcia, E.A. Learning from imbalanced data. *IEEE Trans. Knowl. Data Eng.* **2009**, *21*, 1263–1284.
46. Fernández, A.; García, S.; Herrera, F.; Chawla, N.V. SMOTE for learning from imbalanced data: Progress and challenges, marking the 15-year anniversary. *J. Artif. Intell. Res.* **2018**, *61*, 863–905.
47. Blagus, R.; Lusa, L. SMOTE for high-dimensional class-imbalanced data. *BMC Bioinformatics* **2013**, *14*, 106.
48. Bot, B.M.; Suver, C.; Neto, E.C.; Kellen, M.; Klein, A.; Bare, C.; Doerr, M.; Pratap, A.; Wilbanks, J.; Dorsey, E.R.; et al. The mPower study, Parkinson disease mobile data collected using ResearchKit. *Sci. Data* **2016**, *3*, 160011.
49. Arora, S.; Venkataraman, V.; Zhan, A.; Donohue, S.; Biglan, K.M.; Dorsey, E.R.; Little, M.A. Detecting and monitoring the symptoms of Parkinson's disease using smartphones: A pilot study. *Parkinsonism Relat. Disord.* **2015**, *21*, 650–653.

50. Espay, A.J.; Bonato, P.; Nahab, F.B.; Maetzler, W.; Dean, J.M.; Klucken, J.; Eskofier, B.M.; Merola, A.; Horak, F.; Lang, A.E.; et al. Technology in Parkinson's disease: Challenges and opportunities. *Mov. Disord.* **2016**, *31*, 1272–1282.
51. Dorsey, E.R.; Glidden, A.M.; Holloway, M.R.; Birbeck, G.L.; Schwamm, L.H. Teleneurology and mobile technologies: The future of neurological care. *Nat. Rev. Neurol.* **2018**, *14*, 285–297.
52. Topol, E.J. High-performance medicine: The convergence of human and artificial intelligence. *Nat. Med.* **2019**, *25*, 44–56.
53. Sakar, C.O.; Serbes, G.; Gunduz, A.; Nizam, H.; Sakar, B.E. *Parkinson's Disease Classification*; UCI Machine Learning Repository, 2018. doi:10.24432/C5MS4X.
54. Davis, S.B.; Mermelstein, P. Comparison of parametric representations for monosyllabic word recognition in continuously spoken sentences. *IEEE Trans. Acoust. Speech Signal Process.* **1980**, *28*, 357–366.
55. Selesnick, I.W. Wavelet transform with tunable Q-factor. *IEEE Trans. Signal Process.* **2011**, *59*, 3560–3575.
56. Huang, N.E.; Shen, Z.; Long, S.R.; Wu, M.C.; Shih, H.H.; Zheng, Q.; Yen, N.-C.; Tung, C.C.; Liu, H.H. The empirical mode decomposition and the Hilbert spectrum for nonlinear and non-stationary time series analysis. *Proc. R. Soc. Lond. A* **1998**, *454*, 903–995.
57. Golub, G.H.; Van Loan, C.F. *Matrix Computations*, 4th ed.; Johns Hopkins University Press: Baltimore, MD, USA, 2013.
58. Björck, Å.; Golub, G.H. Numerical methods for computing angles between linear subspaces. *Math. Comput.* **1973**, *27*, 579–594.
59. Chikuse, Y. *Statistics on Special Manifolds*; Springer: New York, NY, USA, 2003.
60. Douzas, G.; Bação, F. Geometric SMOTE: A geometrically enhanced drop-in replacement for SMOTE. *Inf. Sci.* **2019**, *501*, 118–135.
61. Batista, G.E.A.P.A.; Prati, R.C.; Monard, M.C. A study of the behavior of several methods for balancing machine learning training data. *ACM SIGKDD Explor. Newsl.* **2004**, *6*, 20–29.
62. Cortes, C.; Vapnik, V. Support-vector networks. *Mach. Learn.* **1995**, *20*, 273–297.
63. Pedregosa, F.; Varoquaux, G.; Gramfort, A.; Michel, V.; Thirion, B.; Grisel, O.; Blondel, M.; Prettenhofer, P.; Weiss, R.; Dubourg, V.; et al. Scikit-learn: Machine learning in Python. *J. Mach. Learn. Res.* **2011**, *12*, 2825–2830.
64. Brodersen, K.H.; Ong, C.S.; Stephan, K.E.; Buhmann, J.M. The balanced accuracy and its posterior distribution. In Proceedings of the 20th International Conference on Pattern Recognition, Istanbul, Turkey, 23–26 August 2010; pp. 3121–3124.
65. Sokolova, M.; Lapalme, G. A systematic analysis of performance measures for classification tasks. *Inf. Process. Manag.* **2009**, *45*, 427–437.
66. Chicco, D.; Jurman, G. The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation. *BMC Genomics* **2020**, *21*, 6.
67. Fairbanks, G. *Voice and Articulation Drillbook*, 2nd ed.; Harper & Row: New York, NY, USA, 1960.
68. Platt, J.C. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. In *Advances in Large Margin Classifiers*; Smola, A.J., Bartlett, P., Schölkopf, B., Schuurmans, D., Eds.; MIT Press: Cambridge, MA, USA, 1999; pp. 61–74.
69. Guo, C.; Pleiss, G.; Sun, Y.; Weinberger, K.Q. On calibration of modern neural networks. In Proceedings of the 34th International Conference on Machine Learning, Sydney, Australia, 6–11 August 2017; pp. 1321–1330.
70. Vickers, A.J.; Elkin, E.B. Decision curve analysis: A novel method for evaluating prediction models. *Med. Decis. Making* **2006**, *26*, 565–574.
71. Collins, G.S.; Reitsma, J.B.; Altman, D.G.; Moons, K.G.M. Transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD): The TRIPOD statement. *BMJ* **2015**, *350*, g7594.
72. Collins, G.S.; Moons, K.G.M.; Dhiman, P.; Riley, R.D.; Beam, A.L.; Van Calster, B.; Ghassemi, M.; Liu, X.; Reitsma, J.B.; van Smeden, M.; et al. TRIPOD+AI statement: Updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ* **2024**, *385*, e078378.